

BZFlag GPU Renderer Modernization - Tier 3 Report

Tier 3 removes the remaining legacy OpenGL immediate-mode drawing (`glBegin/glEnd`) from BZFlag 2.4 and routes it through the GLBatch utility: vertices accumulate in memory and are drawn with one `glDrawArrays` call per batch instead of one driver call per vertex. Work was done in three user-verified stages after an earlier central-redirect attempt was reverted.

Approach

- Hand-edited conversions per subsystem cluster, one package per stage (the failed Tier 3 attempt used a central macro redirect; this retry avoided it).
- Each stage built, tested in live games (local solo-bot server and public server `bzflag.d03n3r.de`), packaged as a deb, installed, and signed off before the next stage started.
- Every stage committed separately so any stage can be rolled back independently.
- Display-list builders kept immediate-mode on purpose: font glyphs, sun, moon, stars, clouds, mountains, ground lists, rain drops. `glDrawArrays` cannot be captured by display lists; converting them would break rendering.

Conversion coverage

Stage	Subsystem	Blocks	Files
1	HUD / menus (crosshair, tapes, panel, scoreboard, widgets)	33	7
2	World / FX (sky, ground, grid, shots, effects, rain, tracks)	44	10
3	Leftovers (walls, mesh faces, flags, ghost boxes, debug views)	31	13
-	Total converted	97	23 unique files
-	Remain (inside display lists, by design)	15	4 files

Fixes found during testing

- Textured menu items invisible (logo, selector): GLBatch re-enables texcoord/normal client arrays when the batch provides them.

Frame-time regression (~6-7 ms): removed per-flush client-attrib stack push/pop, drew triangle fans natively instead of re-tessellating to triangles on the heap each frame, trimmed attribute restores.

- Tanks untextured until killed: flushes left client-array enable bits dirty; the tank VBO path inherits enable bits, so begin() now snapshots them and end() restores exactly those bits.

- Wall RENDER macros: emit macros re-targeted to the batch; converter-broken backslash continuations repaired.

Measured results (user's machine, same map)

Build	FPS (in-game HUD)	Input lag
Original (pre-Tier-2/3)	~30	baseline
Tier 3 stage 1 (first)	635 (agent-measured)	~7 ms worse
Tier 3 stage 1 (perf-fixed)	667 (agent-measured)	~1 ms
Tier 3 stage 2	671 (agent-measured)	< 3 ms
Tier 3 stage 3 (final)	120+ reported in normal play	< 3 ms

Compatibility

Protocol unchanged (0221). No new libraries. Rendering identical to the pre-Tier-3 build by design; all changes are performance-only on the draw path. Package: bzflag 2.4.31-17 (debs 2.4.31-11 through -17 retained for rollback).